

Simon Chen

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EDUCATION

Boston University

Boston, MA

Bachelor of Arts in Computer Science

Graduated: Spring 2026

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Distributed Systems, Databases.

EXPERIENCE

Software Engineering Intern

February 2026 – June 2026

RESET Standard

Shanghai, China

- Doubled the platform's supported environmental data categories (from 2 to 4) by independently implementing end-to-end water and waste data integration, developing Rails backend ingestion pipelines, PostgreSQL schemas, and client-facing React/Vite/Tailwind dashboards.
- Engineered asynchronous data pipelines with RabbitMQ, Redis, and Sidekiq to process long-running ingestion jobs in the background, decoupling external API calls from user-facing requests.
- Implemented a key client-requested feature by redesigning project-device mapping workflows, refactoring legacy forms to streamline the onboarding of environmental monitoring devices.

Technical Director / Software Engineer

December 2025 – Present

Ezesports

New York, NY

- Migrated an esports league platform from Google Sites to Next.js with a Supabase backend, replacing manual content updates with live scoring and automated season management.
- Designed a PostgreSQL schema on Supabase using Drizzle ORM, supporting 500+ matches across 20+ teams, and built RESTful endpoints for player statistics, standings, and historical data.
- Cut page load time by 75% (from 3.2s to 0.8s) by implementing edge caching on Cloudflare, route-based code splitting, and asset optimization for the Next.js frontend.

IT Consultant

January 2024 – July 2025

Boston University Engineering IT

Boston, MA

- Automated software deployment for 100+ devices using Microsoft Deployment Toolkit and created custom USB deployment solution for Windows 11, reducing deployment time by 40%
- Managed 200+ user accounts and permissions via NAS, conducting access audits and boosting system uptime by 15%
- Enhanced endpoint security with CrowdStrike for 50+ devices, resolving 20+ tickets/week with 100% incident resolution rate

PROJECTS

Persephone | Python, Cython/C++17, Rust, Modal, FAISS

May 2026

- Built and operated a production Kalshi BTC prediction-market trading system that generated \$8K net PnL and traded \$200K volume in its first 4 live days, owning market-data ingestion, theo serving, strategy evaluation, exchange execution, risk gates, and operator controls.
- Engineered a GBM theo layer to replace KDE inference, reducing model query latency from ~1.5ms to ~20μs per query (~75x), while preserving the live theo interface used by strategy evaluation, cache publication, and operator tooling.
- Moved 9 live hot-path kernels into native Cython extensions for order-book maintenance, OBI, RTI, event aggregation, threshold storage, scalar math, and theo post-processing, cutting Kalshi book maintenance from 21.8ms to 195us p50 (112x) and RTI from 2.66ms to 323us p50 (8.3x).
- Built a compiled Cython/C++17 SPSC RTI event ring with cache-line-aligned atomic cursors, power-of-two masked slots, packed uint64 event records, per-venue/pair channels, single-producer enforcement, overflow-to-UNSYNCED recovery, and native ring-to-aggregator drains, benchmarked at 584ns enqueue and 703ns/event drain.

Self-Hosted Media & Application Server | Docker, Linux, Cloudflare, Nginx, Automation

November 2025

- Architected containerized home server infrastructure using Docker Compose, orchestrating 10+ microservices including Sonarr, Radarr, Prowlarr, Jellyfin, Jellyseerr, and qBittorrent for automated media management and streaming with 99%+ uptime
- Implemented secure external access using Cloudflare Tunnels with zero-trust network architecture, eliminating port forwarding vulnerabilities while hosting personal website and services accessible via custom domain

TECHNICAL SKILLS

Languages: C++17, C/C++, Cython, Go, HTML/CSS, Java, JavaScript, Python, Ruby, Rust, SQL, TypeScript

Frameworks & Libraries: Boost Libraries, CuPy, DBSCAN, Django, Drizzle ORM, Express.js, FAISS, FastAPI, Folium, Next.js, Node.js, OpenAI Whisper

Databases & Infrastructure: AsyncStorage, Auth0, CME MDP 3.0, Circom, Cloudflare, Cloudflare Pages, Cloudflare Tunnels, Docker, Docker Compose, Flyway migrations, GeoPandas, Git

Platforms & Systems: AWS, CrowdStrike, Ethereum, Heroku, Kalshi, MetaMask, Microsoft Deployment Toolkit, Modal, NAS, ServiceNow, Thirdweb SDK